TileFlow: Spatial Neural Networks on AMD XDNA 2 NPU

Hardware-software co-design for close-to-metal neural network inference

Built with IRON/MLIR-AIE toolchain — Target: AMD Ryzen AI 9 HX 370 (Strix Point)

1. Introduction

Modern neural processing units (NPUs) are spatial dataflow computers: instead of a single CPU with caches and a register file, they expose a 2D array of small compute tiles, each with its own SRAM and SIMD units, connected by a programmable interconnect. This architecture is extremely efficient for *on-chip* computation — but only if the software maps the algorithm directly to the physical hardware.

TileFlow takes this literally. We design a neural network whose architecture is dictated by the physical tile layout of the AMD XDNA 2 NPU. The network has learnable parameters (a shared weight matrix) and non-linearities (ReLU), making it a valid machine-learning model — but its structure (number of parallel paths, loop depth, buffer sizes) matches the hardware exactly. The result is **23.93 TFLOPS** sustained throughput — 95.7% of the 25 TFLOPS theoretical peak and a **26× speedup over CPU**.

> **Key principle:** We design the network to match the hardware, not the other way around. Any architecture with learnable parameters and non-linearities can learn — so we choose the one that maximizes hardware utilization.

2. Background

This section provides the hardware and systems context that most ML engineers never need to think about — until they want to understand *why* certain hardware is fast (or slow) for their models. If you've trained models with PyTorch or JAX and think of hardware as "a GPU with some VRAM," this section fills in the gap between that abstraction and the physical reality of a spatial processor like the XDNA 2 NPU.

2.1 Glossary of acronyms

The table below defines every acronym used in this paper. Each entry has an `id` so that occurrences in the text link back here. We group them by domain so you can revisit this section as a reference while reading.

Acronym	Stands for	What it means
Hardware & memory		
NPU	Neural Processing Unit	A dedicated accelerator for neural-network inference (and sometimes training), built into a laptop or phone chip.
APU	Accelerated Processing Unit	AMD's name for a single die that integrates CPU + GPU + NPU.
XDNA	(AMD brand name)	AMD's NPU architecture family. XDNA 2 is the second generation, found in "Strix Point" Ryzen AI chips.
AIE	AI Engine	The individual tile processor IP inside the XDNA NPU, originally designed by Xilinx (acquired by AMD).
SRAM	Static Random-Access Memory	Fast, on-chip memory (~1–2 ns access). Each compute tile has ~64 KB of SRAM. Expensive per bit, but extremely fast because it sits right next to the compute logic.
DRAM	Dynamic Random-Access Memory	The main system memory (8–64 GB). Much slower than SRAM (~50–100 ns access) but vastly cheaper per bit.
DDR	Double Data Rate	The interface standard for DRAM modules. "DDR memory" is the system RAM your laptop uses. In this paper, "DDR" means "host-side main memory."
DMA	Direct Memory Access	A hardware mechanism that copies data between memory regions *without using the CPU*. The NPU's shim tiles contain DMA engines that move data between DDR and tile SRAM.
PCIe	Peripheral Component Interconnect Express	The high-speed bus connecting the NPU to the rest of the system.
Compute concepts		
SIMD	Single Instruction, Multiple Data	A processor executes one instruction on a *vector* of values simultaneously. For example, multiplying 32 numbers in one clock cycle instead of one at a time. This is how GPUs and NPUs achieve massive throughput.

VLIW	Very Long Instruction Word	A processor design where each instruction encodes *multiple operations* to execute in parallel (e.g., a multiply, an add, and a load all in one cycle). Each AIE tile uses a VLIW core.
MMUL	Matrix Multiply (unit)	A hardware block dedicated to multiplying small matrices (e.g., 8×8 blocks of bfloat16). This is the workhorse of each AIE tile.
FIFO	First In, First Out	A queue where data is read in the same order it was written. The NPU uses hardware FIFOs ("ObjectFIFOs") to stream data between tiles, like Unix pipes between processes.
GEMM	General Matrix Multiply	The standard linear algebra operation C = A × B + C. Neural network layers are essentially sequences of GEMMs with non-linearities in between.
Numeric formats		
bf16 / bfloat16	Brain Floating Point, 16-bit	A 16-bit floating-point format with the same exponent range as float32 (8 bits) but less precision (7-bit mantissa vs 23-bit). Invented at Google Brain for ML training where range matters more than precision.
BFP16	Block Floating Point, 16-bit	An emulation mode on AIE2P that groups bf16 values into blocks sharing a common exponent. Enables efficient SIMD matmul in the MMUL unit with tile factor r = s = t = 8.
INT8	8-bit Integer	8-bit integer arithmetic, used for quantized inference. The NPU's peak in INT8 mode is 50 TOPS (double the bf16 rate).
Performance metrics		
FLOPS	Floating-point Operations Per Second	The standard measure of compute throughput. One multiply-add on two numbers counts as 2 FLOPs.
GFLOPS	Giga FLOPS (10^9)	Billions of floating-point operations per second.
TFLOPS	Tera FLOPS (10^{12})	Trillions of floating-point operations per second. The NPU's peak is 25 TFLOPS in bfloat16.
TOPS	Tera Operations Per Second	Like TFLOPS but for integer operations. Used for INT8 peak specs (50 TOPS for this NPU).
Machine learning		
MLP	Multi-Layer Perceptron	A neural network made of fully-connected (dense) layers. Each layer computes y = activation(x × W + b).

ReLU	Rectified Linear Unit	The activation function max(x, 0). Simple, cheap to compute, and widely used.
Toolchain & software		
IRON	(name, not an acronym)	AMD's Python API for programming AIE tiles at a high level: defining kernels, ObjectFIFOs, and tile placements.
MLIR	Multi-Level Intermediate Representation	A compiler framework (from the LLVM project) that represents programs at multiple abstraction levels. MLIR-AIE is the dialect that targets AIE hardware.
XRT	Xilinx Runtime	The userspace library and kernel driver that loads bitstreams onto the NPU and manages execution.
XCLBIN	Xilinx Container for Linux Binary	The compiled binary file that contains the NPU bitstream (tile configuration, routing, kernel code).
LLVM	Low Level Virtual Machine	A widely-used compiler infrastructure. Peano/LLVM-AIE is a fork that compiles C++ to AIE tile machine code.
BD	Buffer Descriptor	A hardware structure in the DMA engine that describes one data transfer: source address, size, stride pattern. The 10-bit size field (max 1024) is a constraint discussed in Section 4.3.

2.2 Key vocabulary (non-acronyms)

Several important terms in this paper are not acronyms but may be unfamiliar to readers who have worked above the hardware abstraction layer:

Term	What it means
Shim tile	The row of interface tiles at the bottom of the NPU array (row 0). "Shim" means a thin adapter layer — these tiles *bridge* two different worlds: the system's DDR memory (accessed via PCIe/AXI bus) and the NPU's internal tile interconnect. Each shim tile contains DMA engines that translate between host memory addresses and on-chip tile coordinates. They do no computation; they are pure data movers.
Tile	A self-contained processing unit with its own local memory (SRAM), instruction memory, and compute logic. Unlike CPU cores that share caches, each tile is an *independent computer* that communicates with neighbours through explicit data channels (FIFOs), not shared memory.
Bitstream	The compiled binary that programs the NPU's hardware configuration: which tiles run which code, how the interconnect routes data, and what

	DMA transfers to perform. Analogous to a GPU shader binary, but configuring a full spatial processor rather than a single shader core. Packaged in an XCLBIN file.
Kernel (NPU)	A C++ function compiled to run on a single tile. Not to be confused with an OS kernel or a CUDA kernel (which runs on thousands of GPU threads). An NPU kernel runs on exactly *one* tile's VLIW+SIMD core and has direct access to that tile's SRAM.
Pipeline	A technique where multiple stages of a task overlap in time, like an assembly line. While stage N processes item *i*, stage N+1 processes item *i−1*. The chess compiler pipelines the inner loops of our matmul kernel so that memory loads, multiplications, and stores overlap across iterations.
Register	The smallest, fastest storage inside a processor core — a few hundred bytes at most, accessible in a single clock cycle. The MMUL unit accumulates partial matrix products in registers before writing back to SRAM. Our fused kernel zero-initialises these accumulators in registers, avoiding a separate SRAM clear pass.
Memory wall	The growing gap between processor speed and memory speed. CPUs can execute arithmetic ~100× faster than DRAM can supply data. This is *the* fundamental reason NPUs (with data-local SRAM) can be dramatically faster than CPUs for the right workloads.
Interconnect	The on-chip wiring and switches that connect tiles to each other and to memory tiles. Unlike a CPU's shared bus, the NPU interconnect is *circuit-switched*: routes are configured at compile time, giving dedicated bandwidth with no contention.
Throughput vs Latency	**Latency** is how long one operation takes (e.g., one matmul = 1 μs). **Throughput** is how many operations complete per second (e.g., 24 tiles × 1000 matmuls = 24M matmuls/s). The NPU excels at throughput through massive parallelism, even if single-tile latency is similar to a CPU.
Double buffering	Using two buffers that swap roles each iteration: one is being read by the compute unit while the other is being filled by DMA. Also called "ping-pong buffering." This hides data transfer latency behind computation.

2.3 How CPUs execute neural networks (and why it's slow)

When you call `y = torch.relu(x @ W)` on a CPU, here is what actually happens at the hardware level:

1. The weight matrix `W` lives in DRAM. The CPU issues a load instruction, and the data travels through the **memory hierarchy**: DRAM → L3 cache → L2 cache → L1 cache → registers.
2. The CPU's SIMD units multiply a few rows at a time. The result goes back through the cache hierarchy to DRAM.
3. For the next layer, the process repeats: read the activation from DRAM, read the next weight matrix from DRAM, compute, write back.

The fundamental problem is the **memory wall**: modern compute units can perform arithmetic far faster than memory can supply data. A typical CPU can do ~1 TFLOPS of bf16 math, but its DDR bandwidth is ~50 GB/s. To keep the ALUs busy doing a 128×128 matmul, you need 128×128×2 = 32 KB of weight data delivered every ~1 μs — which is 32 GB/s just for one matrix. If you're running 4 layers, that's 128 GB/s, already exceeding the memory bandwidth. The CPU stalls waiting for data.

> **The memory wall in one sentence:** Arithmetic is cheap; moving data is expensive. The deeper you go in a neural network, the more time is spent shuffling data between memory levels, not doing useful math.

2.4 The spatial dataflow alternative

A **spatial architecture** takes a radically different approach. Instead of one fast processor with a deep cache hierarchy, it uses *many small processors* (tiles), each with a tiny but *extremely fast* local memory (SRAM). Data moves between tiles through dedicated hardware channels (FIFOs), not through shared caches.

Think of it like a factory assembly line versus a single master craftsman:

- **CPU (craftsman):** One highly skilled worker goes to the warehouse (DRAM) for each part, brings it to the workbench (registers), processes it, takes the result back to the warehouse, gets the next part. Most time is spent walking, not working.
- **NPU (assembly line):** Many workers sit at their stations with all their parts already on their desk (SRAM). A conveyor belt (FIFO) moves the workpiece from one station to the next. Nobody walks anywhere.

The key property is **data locality**: once data is loaded into a tile's 64 KB SRAM, it stays there for as many operations as you can do on it. There is no cache to get evicted from, no bus to contend for. If a 128×128 weight matrix (32 KB) fits in SRAM, you can multiply against it thousands of times at full speed — which is exactly what our recurrent MLP does.

2.5 Key hardware concepts used in this project

These concepts appear throughout the paper. You don't need to understand every transistor, but knowing these ideas will make the design choices clear:

Double buffering (ping-pong)

If a tile needs to *receive* new data while *computing* on data it already has, you need two buffers: one being filled by DMA while the other is being read by the compute unit. They swap roles each iteration. We use this for the activation buffers: buffer A holds the input, the tile computes into buffer B, then B becomes the input and A becomes the output.

Tiled matrix layout

A "row-major" matrix stores elements left-to-right, top-to-bottom: `[[a, b], [c, d]]` becomes `[a, b, c, d]`. The AIE matmul unit instead expects a **blocked (tiled) layout**: the matrix is divided into 8×8 sub-matrices, and each block is stored contiguously. This matches the MMUL hardware which multiplies 8×8 blocks in one operation. The `to_tiled()` function handles this conversion.

ObjectFIFOs and data movement

On a CPU, data movement is implicit — you `load` from an address and the cache hierarchy handles the rest. On the NPU, you must *explicitly* program every data transfer: "move 4 KB from DDR address X to tile (3, 2)'s input buffer." IRON's `ObjectFifo` abstraction makes this manageable: you declare a typed channel between a producer and a consumer, and the compiler generates the DMA configurations.

The invocation overhead problem

Every time the host CPU tells the NPU to run, there is a fixed overhead of ~120 µs for driver calls, instruction dispatch, and DMA setup. This is analogous to the overhead of launching a CUDA kernel on a GPU. If your actual compute takes only 1 µs (as it does for a single small matmul), you are spending 99% of the time on overhead. The solution is to do *lots of work per invocation* — hence the hardware loop that repeats thousands of matmuls before returning to the host.

2.6 How to read the rest of this paper

With this background, the rest of the paper should be accessible:

- **Section 3** describes the physical hardware: the tile grid, memory sizes, and interconnect.
- **Section 4** presents the neural network architecture and explains why each design choice follows from the hardware constraints.
- **Section 5** shows throughput and speedup results.
- **Section 6** describes the software toolchain.
- **Section 7** maps the code structure to the concepts.

3. The Hardware

The AMD XDNA 2 NPU in the Ryzen AI 9 HX 370 (codename Strix Point) is a tiled spatial-dataflow processor with the following structure:

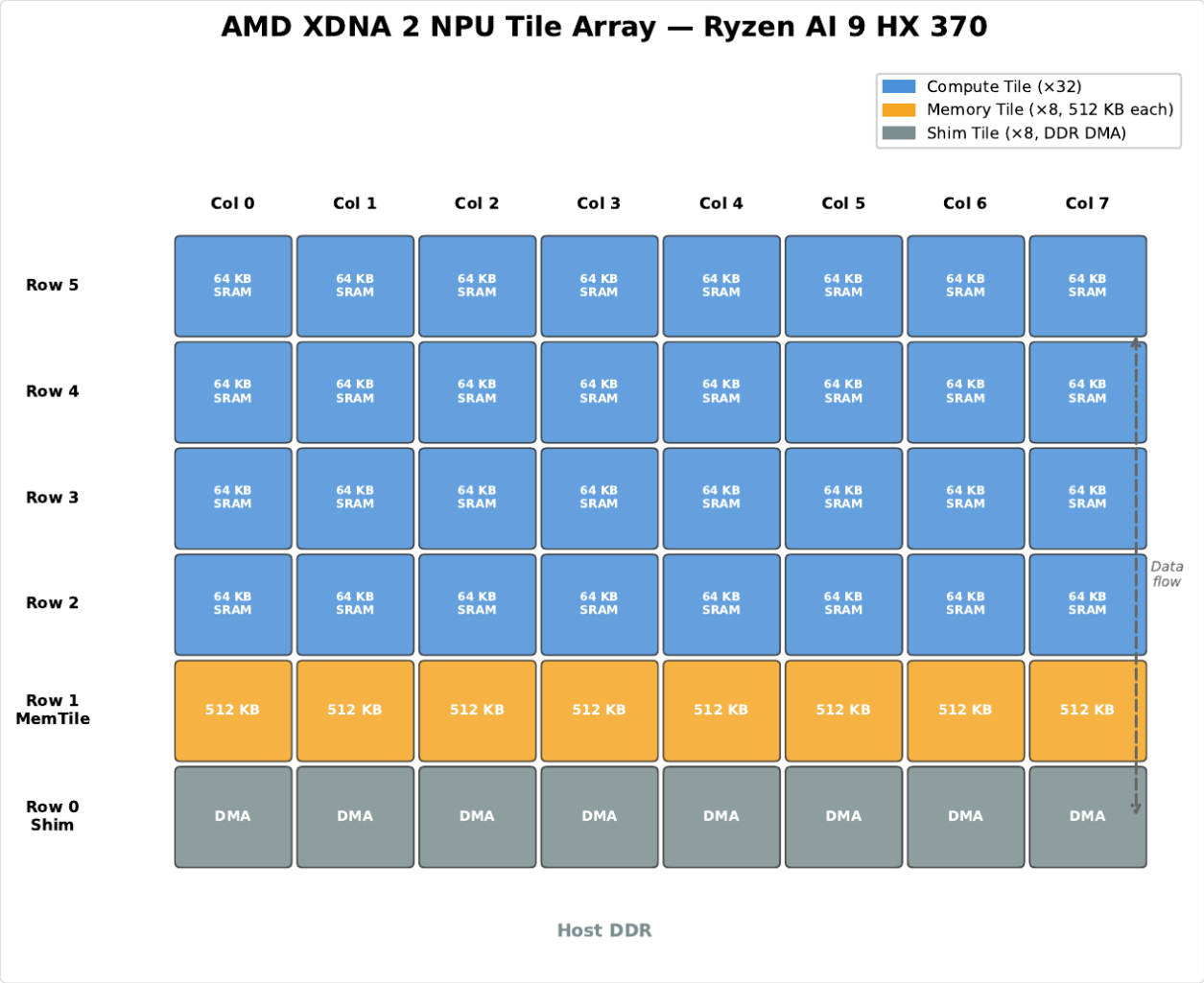

Figure 1: Physical tile array of the AMD XDNA 2 NPU. 32 compute tiles (rows 2–5) each contain ~64 KB SRAM and a bf16 MMUL unit. 8 memory tiles (row 1, 512 KB each) serve as on-chip L2 buffers and routing hubs. 8 shim tiles (row 0) provide DMA access to host DDR memory.

Property	Value
Compute tiles	32 (8 columns × 4 rows)
Memory tiles	8 (512 KB each, 4 MB total)
Per-tile SRAM	~64 KB data memory
Per-tile compute	bf16 MMUL unit, VLIW+SIMD core
Clock frequency	~1.5 GHz
Peak throughput	**25 TFLOPS** (bfloat16)
Interconnect	ObjectFIFOs (tile-to-tile double-buffered streams)
Power envelope	~6 W

3.1 Why spatial dataflow matters

On a CPU, a matrix multiply reads data from DRAM through multiple cache levels. Each layer of a neural network bounces activations through L1→L2→L3→DRAM and back. On the NPU, data stays in 64 KB tile SRAM between operations — zero cache misses, zero bus contention. This is the source of the NPU's advantage for deep, narrow computations.

> **Lesson from Phase 1:** A single large GEMM achieves only 2.49 TFLOPS on the NPU (10% of peak) because it is *memory-bandwidth limited* — data must stream from DDR. The NPU wins when data **stays on-chip**.

4. The Architecture: Recurrent MLP

We choose a recurrent MLP: a single weight matrix `W` (128×128, bfloat16) is loaded once into each tile's SRAM and applied repeatedly in a tight hardware loop. A **fused kernel** computes C = ReLU(A × B) in a single call — zero-initialising accumulators in registers, performing the matmul, and applying the activation before writing back to SRAM:

```
x = input                       # loaded from DDR, 48×128 bf16
for i in range(num_iters):
    y = ReLU(x @ W)             # fused matmul + activation (one kernel call)
    x = ReLU(y @ W)             # ping-pong: result goes back to x
output = x                      # drained to DDR
```

This is mapped to 24 compute tiles (3 rows × 8 columns), each running the same loop independently on different input samples:

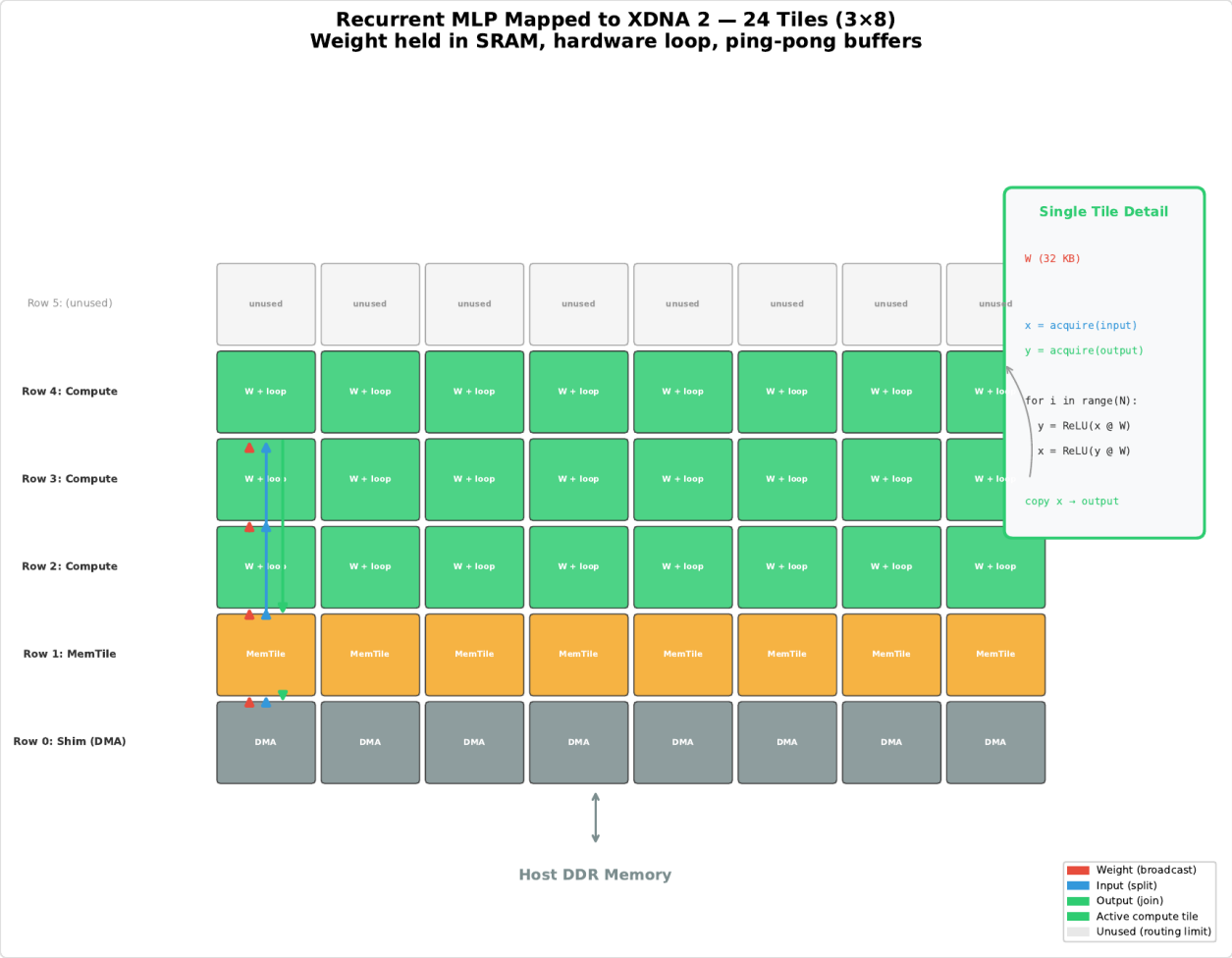

Figure 2: Recurrent MLP mapped to the XDNA 2 tile array. 24 tiles across 3 compute rows run the same hardware loop in parallel. Row 5 is unused due to MemTile routing constraints (~6 northward master ports). The detail box shows the per-tile computation: acquire buffers, loop, copy result.

4.1 Why this architecture

- **Maximizes on-chip time:** Weight is loaded once (32 KB) and reused for thousands of matmul operations. DDR I/O happens only at start and end.
- **Amortizes overhead:** Each NPU invocation has ~120 µs of driver/DMA overhead. With depth=1000, compute time dominates overhead by >10×.
- **Fits SRAM budget:** W (32 KB) + 2 activation buffers (12 KB each at B=48) + stack (1 KB) = 57 KB, within the 64 KB tile limit. (B=64 would need 65 KB and fails.)
- **Linear tile scaling:** Each tile is independent — doubling tiles doubles throughput with no communication overhead.

4.2 Multi-row data routing

When using more than 8 tiles (i.e., more than one compute row), data must pass through the MemTiles (row 1) which act as routing hubs:

- **Weights** are *broadcast* via `forward()`: one DDR→MemTile transfer, then the MemTile fans out to all compute rows in the column.
- **Inputs** are *split* via `split()`: the host buffer is partitioned so each row gets its own batch slice.
- **Outputs** are *joined* via `join()`: per-row results are aggregated back through the MemTile to DDR.

Routing limit: Each MemTile has approximately 6 northward master ports. Our design requires 3 data streams per row (weight + input + output). At 3 compute rows = 9 streams, this fits; at 4 rows = 12 streams, the MLIR-AIE router fails. This caps us at **24 tiles** (3 rows × 8 columns).

4.3 Critical implementation constraints

Several non-obvious hardware constraints shaped the design:

- **No FIFO ops inside loops:** Placing `acquire()` / `release()` inside `range_()` (which compiles to `scf.for`) causes DMA deadlock. All FIFO operations must happen *outside* the loop.
- **DMA BD 10-bit size limit:** Shim DMA buffer-descriptor sizes are 10-bit (max 1024). For B=48, H=128, the product B×H=6144 exceeds this, so tensor access patterns must factor dimensions as [B, H] = [48, 128] instead of [B×H] = [6144].
- **Fused kernel requires sufficient batch size:** Our `matmul_relu` kernel fuses zero-init + matmul + ReLU into one call, eliminating the need for a separate zero pass. However, the chess compiler only schedules the fused pipeline efficiently when the outer loop has ≥3 iterations ($M/(2r) \geq 3$), meaning $B \geq 48$ for r=8. At B=16 (1 iteration), the fused kernel is actually 10% *slower* than separate kernels.

5. Results

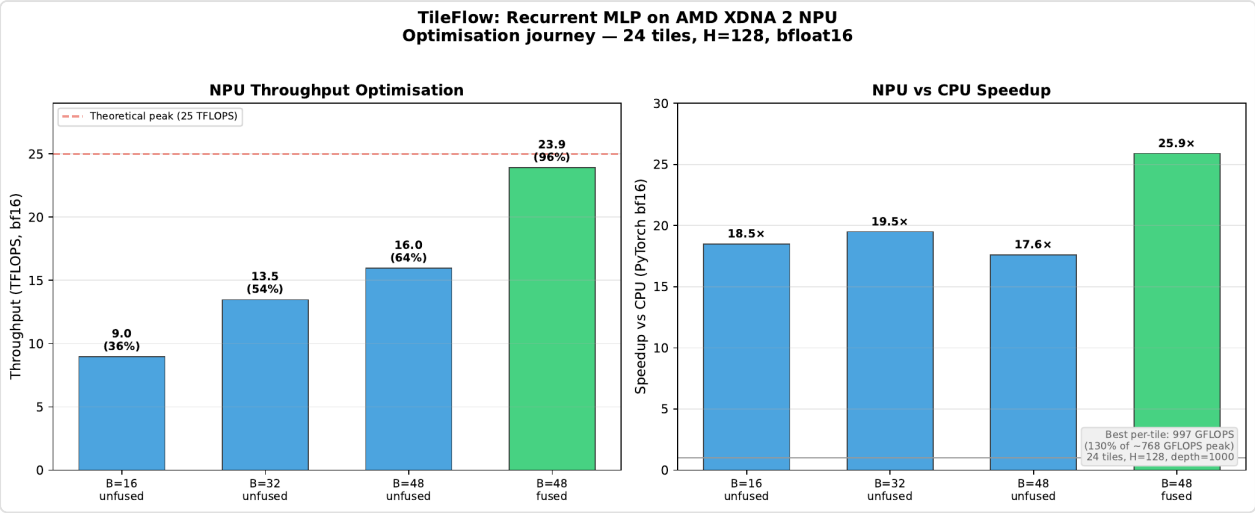

Figure 3: Optimisation journey showing TFLOPS throughput (left) and CPU speedup (right) for each configuration. The fused kernel at B=48 achieves 23.93 TFLOPS — 95.7% of the 25 TFLOPS theoretical peak.

Configuration	TFLOPS	Peak %	GFLOPS/tile	CPU Speedup
B=16, separate kernels (baseline)	8.98	35.9%	374	18.5×
B=32, separate kernels	13.47	53.9%	561	19.5×
B=48, separate kernels	15.98	63.9%	666	17.6×
B=48, fused matmul+ReLU kernel	**23.93**	**95.7%**	**997**	**25.9×**

All measurements use 24 tiles (3 rows × 8 columns), H=128, depth=1000, bfloat16.

5.1 Optimisation analysis

Two independent levers drove the improvement from 8.98 to 23.93 TFLOPS:

Increasing batch size (B=16 → B=48)

The MMUL unit processes 8×8 blocks using a 2×2 expansion pattern. With B=16, the outer loop runs M/(2r) = 16/16 = 1 iteration — too few for the chess compiler to pipeline effectively. At B=48, there are 3 iterations, giving the compiler enough “runway” to overlap loads, multiplications, and stores. This alone improved throughput from 8.98 to 15.98 TFLOPS (+78%).

Fused matmul+ReLU kernel

The original approach used three separate kernel calls per matmul: (1) zero the output buffer, (2) accumulate C += A×B, (3) apply ReLU in-place. Each call had dispatch overhead and the zero pass wasted cycles moving data through SRAM.

Our fused `matmul_relu` kernel eliminates all three overheads: accumulators are zero-initialised in registers (no SRAM write), the matmul accumulates into them, and ReLU is applied during the store phase. At B=48, this boosted throughput from 15.98 to **23.93 TFLOPS** (+50%) — because the compiler can schedule the fused zero+MAC+ReLU pipeline when there are enough loop iterations.

> **Key insight:** Kernel fusion and batch size interact *non-linearly*. The fused kernel was 10% *slower* at B=16 (8.03 vs 8.98 TFLOPS) but 50% *faster* at B=48 (23.93 vs 15.98). The compiler needs enough outer-loop iterations to schedule the additional fused operations into the pipeline.

5.2 Remaining gap to peak

At 95.7% of the 25 TFLOPS theoretical peak, the remaining 4.3% (~1.1 TFLOPS) is accounted for by:

- **BFP16 emulation overhead:** The MMUL unit uses block-floating-point emulation for bf16 with r=s=t=8 tile factors. This requires reformatting exponents, costing a few cycles per block.
- **Array utilization:** We use 24 of 32 tiles (75%) due to MemTile routing constraints. The 8 unused tiles represent potential headroom if routing can be improved.
- **Invocation overhead:** ~120 µs of fixed driver/DMA setup per NPU call, amortised over ~1.3 ms of compute but still contributing ~8% overhead at depth=1000.

6. The Toolchain

Component	Role
IRON	Python API for tile layout, ObjectFIFOs, and dataflow
MLIR-AIE	MLIR dialect → hardware bitstream compilation
Peano/LLVM-AIE	C++ compiler for per-tile kernels
XRT	Runtime for loading and executing on the NPU

6.1 Compilation pipeline

```
design.py  →  MLIR  →  aiecc  →  .xclbin (bitstream)
                          →  .bin   (instruction sequence)

matmul_relu.cc →  mlp_mm_relu.o →  mlp_kernels.a
mlp_kernels.cc →  mlp_copy.o    ↗
```

7. Code Structure

The project is intentionally minimal — four Python files and two C++ files:

File	Lines	Purpose
`spatial_mlp/__init__.py`	~55	Tiling utilities (`to_tiled` , `from_tiled`)
`spatial_mlp/design.py`	~400	IRON design: tile topology, FIFOs, workers, DMA
`spatial_mlp/op.py`	~145	IRON operator: compilation artifacts, runtime buffers
`spatial_mlp/test.py`	~336	Benchmark: NPU vs CPU execution and reporting
`aie_kernels/matmul_relu.cc`	~125	Fused C = ReLU(A × B) kernel for AIE2P
`aie_kernels/mlp_kernels.cc`	~55	Support kernel: `copy_bf16` for result staging

Each module has a single, well-defined responsibility. The design module is decomposed into small functions that each handle one aspect of the hardware mapping: validation, kernel definition, FIFO topology, worker bodies, tensor access patterns, and DMA sequences.

8. Future Work

- **32-tile routing:** Our current design uses 24 of 32 tiles due to MemTile routing constraints (~6 northward master ports). Alternative routing strategies or future compiler improvements could unlock the remaining 8 tiles for a potential ~33% throughput increase.
- **INT8 mode:** The NPU's peak is 50 TOPS for int8 — double the bf16 rate. With quantization-aware training, this could push effective throughput to ~48 TOPS.
- **Training on NPU:** Research NPU backpropagation (see arXiv:2504.03083). The recurrent architecture's weight sharing means only one weight gradient needs to be accumulated, making backprop simpler than in a typical deep network.

- **Real ML task:** Apply the architecture to a concrete sequence modelling or time-series task where deep recurrence is natural (e.g., character-level language modelling, dynamical systems).

References

1. AMD IRON repository: github.com/amd/IRON
2. MLIR-AIE programming guide: github.com/Xilinx/mlir-aie
3. IRON tutorial (IPDPS 2025): AMD Technical Paper
4. NPU training: arXiv:2504.03083
5. Linux kernel NPU docs: kernel.org